



Orreus: An Interactive Orrery Web App

VISUALIZING THE SOLAR SYSTEM AND BEYOND

TEAM NAME: THE DOMIRA: THE GUARDIANS OF AURCTIA

CHALLENGE: CREATE AN ORRERY WEB APP THAT DISPLAYS NEAR-EARTH OBJECTS

TEAM MEMBERS:
RAGHUVARAN D



What is an Orrery?

AN ORRERY IS A MECHANICAL MODEL OF THE SOLAR SYSTEM. THE NAME ORIGINATES FROM CHARLES BOYLE, TO WHOM A MECHANICAL MODEL WAS PRESENTED IN 1713. OUR GOAL WAS TO MODERNIZE THIS CONCEPT WITH TODAY'S TECHNOLOGY





We Challenge The Subject

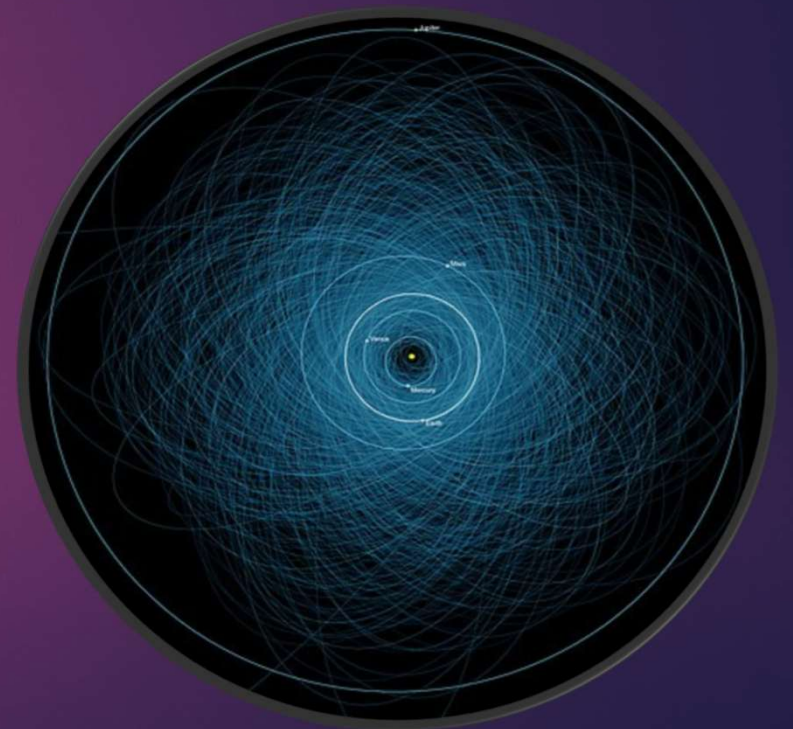
THE SOLAR SYSTEM IS VAST, AND WITH INCREASING INTEREST IN SPACE EXPLORATION, IT'S CRUCIAL TO CREATE INTERACTIVE TOOLS TO EDUCATE THE PUBLIC AND PROVIDE DETAILED INFORMATION ABOUT SPACE, ESPECIALLY CONCERNING NEAR-EARTH OBJECTS LIKE ASTEROIDS AND COMETS





Introducing Orreus

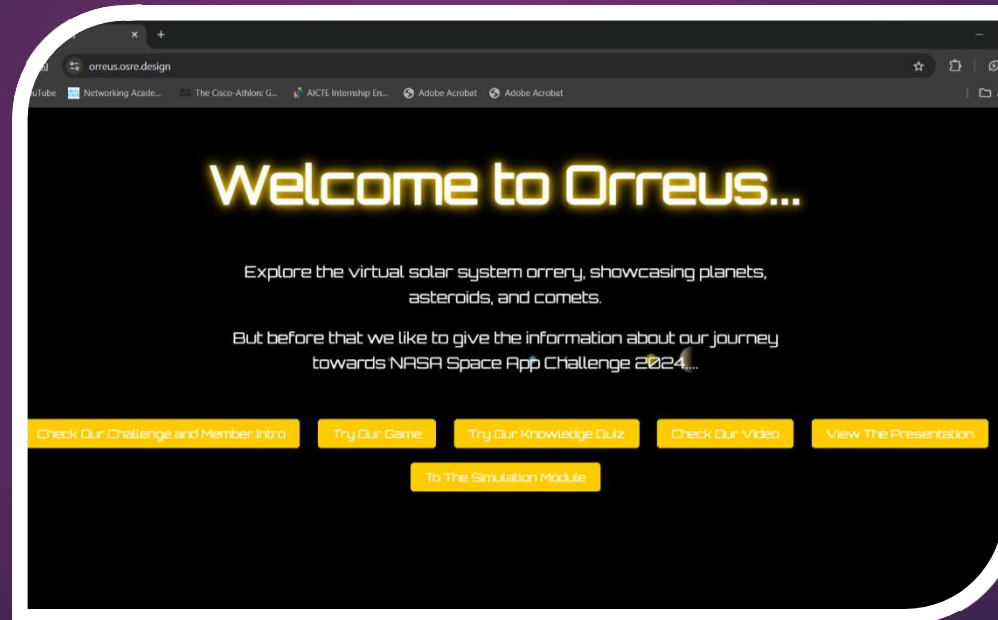
ORREUS IS A VIRTUAL ORRERY, DESIGNED TO MAKE LEARNING ABOUT THE SOLAR SYSTEM IMMERSIVE. IT DISPLAYS CELESTIAL BODIES, INCLUDING PLANETS, NEAR-EARTH ASTEROIDS, COMETS, AND PHAS. THE APP ENABLES USERS TO INTERACT WITH REAL-TIME DATA AND EXPLORE MISSIONS.





Our Application

Front Page



Our Application

Simulation of
Orrery



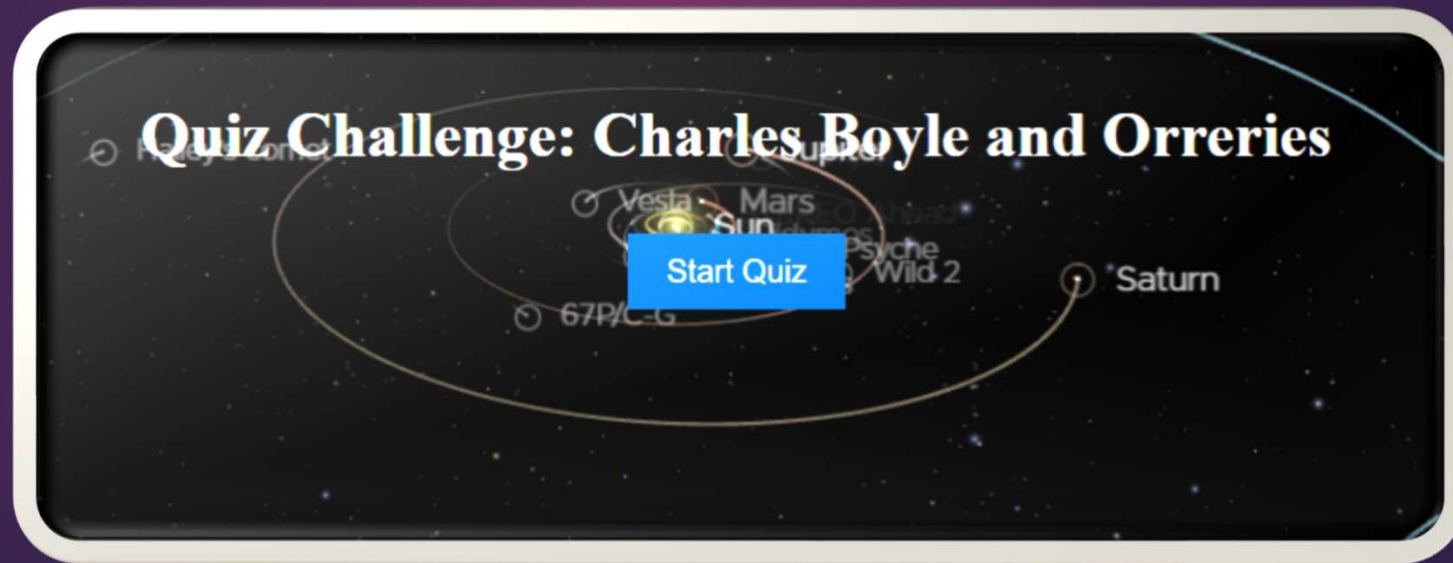
Our Application

Simulation of
Game



Our Application

Quiz Simulation





How Users Interact with Orreus?

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Orreus as an Educational Tool

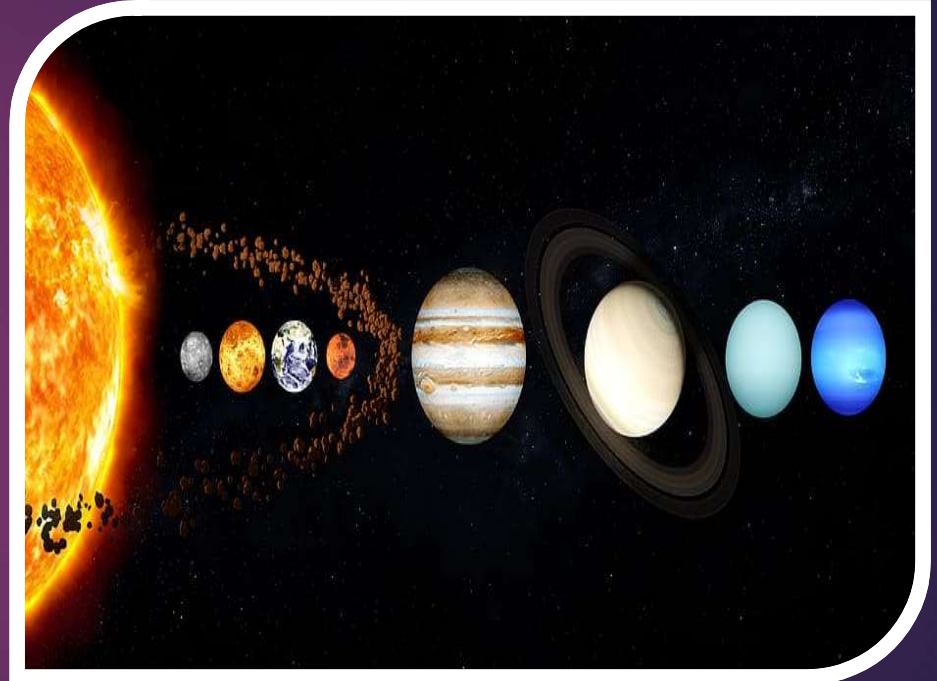
ORREUS IS MORE THAN JUST A TOOL FOR EXPLORATION—IT'S A POWERFUL EDUCATIONAL PLATFORM THAT CAN BE USED IN CLASSROOMS AND MUSEUMS TO TEACH STUDENTS ABOUT THE SOLAR SYSTEM AND SPACE SCIENCE.





Looking Forward

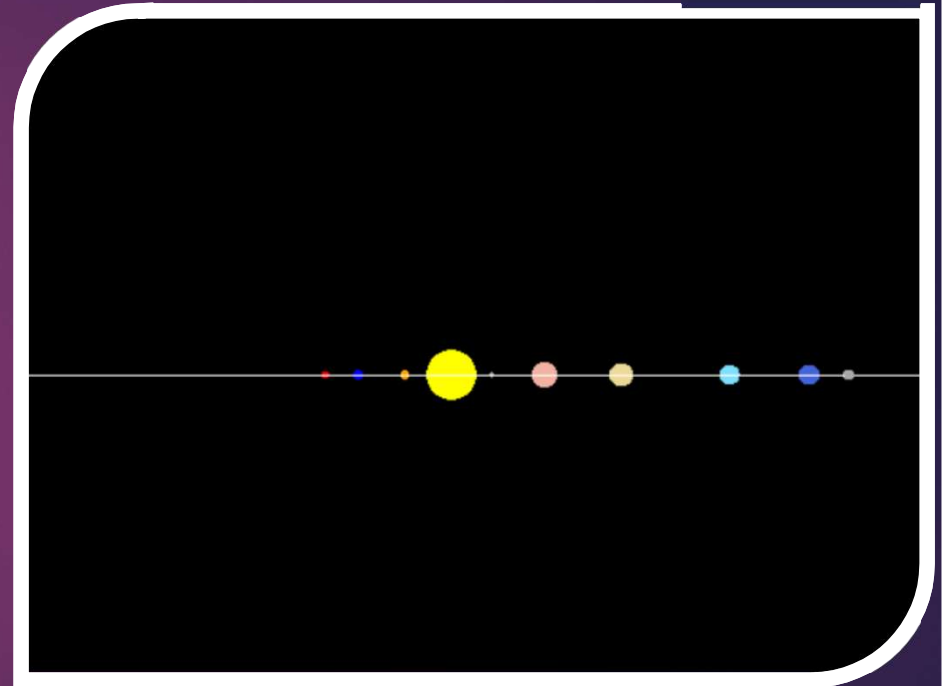
LOOKING AHEAD, WE HAVE PLANS TO ENHANCE ORREUS FURTHER BY INTEGRATING VR EXPERIENCES, ALLOWING USERS TO PARTICIPATE IN ASTEROID MINING MISSIONS AND EXPLORE EXOPLANETARY SYSTEMS.





Try Orreus Today

WE INVITE YOU TO EXPLORE ORREUS TODAY! WHETHER YOU'RE A SPACE ENTHUSIAST OR AN EDUCATOR, OUR PLATFORM OFFERS SOMETHING EXCITING FOR EVERYONE. THANK YOU FOR YOUR TIME, AND WE'RE HAPPY TO ANSWER ANY QUESTIONS.





Reference

MODELS:

[HTTPS://CODEPEN.IO/JCOULTTERDESIGN/PEN/ZXXBEP?EDITORS=1111](https://codepen.io/jcoultterdesign/pen/zxxbep?editors=1111)

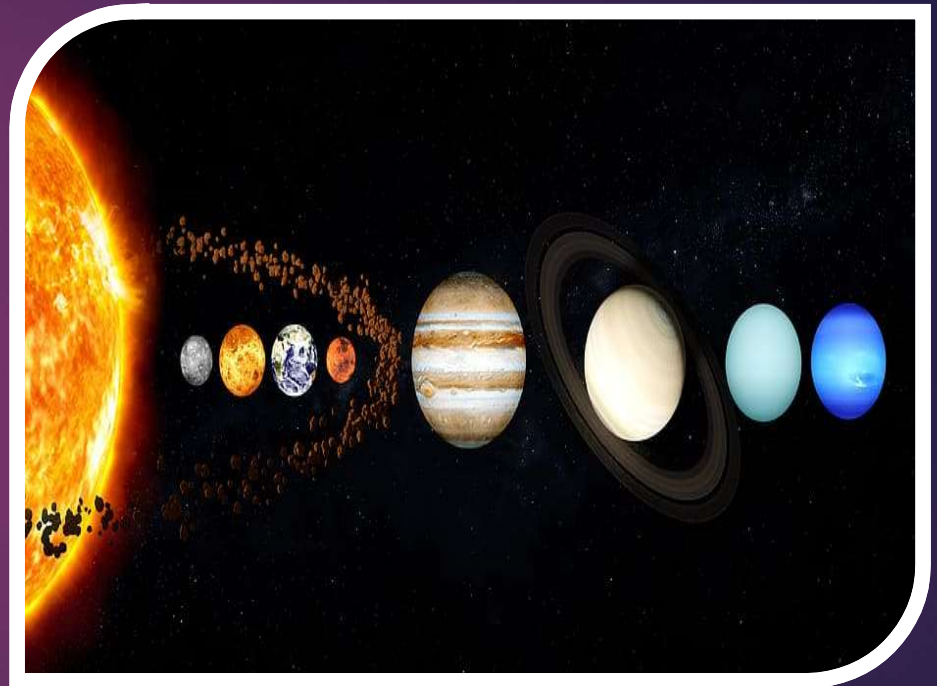
[HTTPS://CODEPEN.IO/WILLEMCRNLSSN/PEN/NABXOO](https://codepen.io/willemcrlssn/pen/nabxoo)

[HTTPS://DATA.NASA.GOV/SPACE-SCIENCE/NEAR-EARTH-COMETS-ORBITAL-ELEMENTS-API/YSQN-VD8V/ABOUT_DATA](https://data.nasa.gov/space-science/near-earth-comets-orbital-elements-api/ysqn-vd8v/about_data)

THEORY:

RESOURCES OF OUR CHALLENGE

[HTTPS://WWW.SPACEAPPSCHALLENGE.ORG/NASA-SPACE-APPS-2024/CHALLENGES/CREATE-AN-ORRERY-WEB-APP-THAT-DISPLAYS-NEAR-EARTH-OBJECTS/?TAB=RESOURCES](https://www.spaceappschallenge.org/nasa-space-apps-2024/challenges/create-an-orrery-web-app-that-displays-near-earth-objects/?tab=resources)





Project Codes and Videos

NASA SPACE APP CHALLENGE 2024 PROJECT LINK:

[HTTPS://GITHUB.COM/RAGHUVARANDBECSE32/IORRERA](https://github.com/RAGHUVARANDBECSE32/IORRERA)

PPT LINK:

[HTTPS://1DRV.MS/P/C/1375A2F2B81AEE6F/EYQBEXLC9ZT-HPX1D-ZVH9ZKBUBUO8JRSEKLOVD04S4QVHA?E=POJYSG](https://1drv.ms/p/c/1375A2F2B81AEE6F/EYQBEXLC9ZT-HPX1D-ZVH9ZKBUBUO8JRSEKLOVD04S4QVHA?e=POJYSG)

WEBSITE:

[HTTPS://LGN.ORREUS.OSRE.DESIGN/](https://lgn.orreus.osre.design/)





Thank You